

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Easy

Question

$k^2 - 53 = 91$

What is the positive solution to the given equation?

Answer

- A. 144
- B. 72
- C. 38
- D. 12

Correct Answer: D

Rationale

Choice D is correct. Adding 53 to each side of the given equation yields $k^2 = 144$. Taking the square root of each side of this equation yields $k = \pm 12$. Therefore, the positive solution to the given equation is 12.

Choice A is incorrect. This is the positive solution to the equation $k^2 - 53 = 20,683$, not $k^2 - 53 = 91$.

Choice B is incorrect. This is the positive solution to the equation $k^2 - 53 = 5,131$, not $k^2 - 53 = 91$.

Choice C is incorrect. This is the positive solution to the equation $k^2 - 53 = 1,391$, not $k^2 - 53 = 91$.